Hwanseok Sim

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RESEARCH INTERESTS

- Develop deep learning methods to interpret genetic variation and understand the regulatory logic of the human genome.
- Uncover the genetic and regulatory architecture of autism and other neurodevelopmental disorders through cohort scale genomic analysis.
- Advance the interpretability of genomic models to connect learned representations with biologically meaningful mechanisms.

EDUCATION

Korea University (GPA: 4.35/4.5) — Seoul, Republic of Korea
B.Sc. in Computer Science and Engineering — 2020 – present
B.Sc. in Biosystem and Biomedical Sciences — 2020 – present

Daegu Science High School for the Gifted — Daegu, Republic of Korea
Science-gifted secondary program — 2017 – 2020

RESEARCH EXPERIENCE

Lab of Human Genomics (joonanlab.github.io), Korea University — 2024 – present
Supervisor: Prof. Joon-Yong An

- Developed an annotation-aware genomic representation framework using knowledge distillation, improving the variant effect prediction performance of a semi-supervised genomic model.
- Performed *in silico* MPRA on psychiatric-disorder GWAS variants to nominate candidate functional variants for experimental validation.
- Developed models to analyze ASD subtypes and predict phenotypes using whole genome *de novo* variants from large scale ASD cohorts (SSC, MSSNG, SFARI, Korean).
- Designed cell-type-specific synthetic enhancers targeting the MEF2C gene, which shows ASD-specific regulatory patterns, using deep learning models.
- Contributed to the development of a brain organoid *in silico* perturbation framework (AIVC); analyzed gene–gene networks using network propagation.

Lab of Neuronal Cell Death, DGIST — 2019 (9 months)
Supervisor: Prof. Seong-Woon Yu

Nanotechnology Group, Columbia University — 2018 (2 weeks)
Supervisor: Prof. Glen Kowach

Inorganic Nanomaterials Laboratory, Yeungnam University — 2017 (5 months)
Supervisor: Prof. Misook Kang

PUBLICATIONS

(co-first authors, † corresponding authors)*

First-author Publications

1. **Sim H***, Kim Y, Kim SW, Ryu Y, An JY†, annDNA: Learning Annotation-Aware Genomic Representations via Knowledge Distillation *(ICML GenBio Spotlight, 2026)*
2. **Sim H***, Kim Y*, Park W, Park S, Nam JW, An JY†, Genomic Language Models: Architecture, Variant Interpretation, and the Context Distribution Problem *(Under review, Experimental & Molecular Medicine)*
3. **Sim H***, Kim Y, An JY†, In Silico Mutagenesis Using Genomic Foundation Models Identifies Functional Variants and Cell-Type-Specific Regulation at the CD19 Enhancer *(Manuscript in preparation)*

Other Publications

4. Koh IG*, Chang E*, Choi YS*, Kim SW, Kim Y, Lee H, Byeon G, Ryu Y, Kim S, Lee J, Park H, **Sim H**, Ryu Y, Shim W, Lee J, Bautista Salazar N, Mendes M, Engchuan W, Zhou X, Son JH, Lee J, Bong G, Kim IB, Han JH, Werling D, Kim SH, Oh M, Kim MS, Lee D, Kim J, Lee YS, Sun W, Kim E, Scherer S, Jeon M†, Yoo HJ†, An JY†, AI-driven framework modeling perturbation in brain organoids reveals candidate genes for autism *(Under review, Cell)*
5. Kim Y*, Lee H*, Hong H*, **Sim H**, Kim SW, Son JH, Bong G, Han JH, Kim SH, Lee J, Kim E, Oh M, Kim IB, Yoo HJ†, An JY†, Modeling germline mutational mechanisms to generate synthetic de novo variants *(Under review, eBioMedicine)*
6. Kim Y*, Kim IB, Kim SW, Lee H, Koh IG, Choi S, Ryu Y, Chang E, Kim C, **Sim H**, Min CW, Jeong HE, Byun H, Lee J, Han JH, Oh M, Kim SH, Kim E, Jeon M†, Yoo HJ†, An JY†, Deep learning reveals autism subtypes defined by distinct regulatory architectures and variant burden *(Under review, Nature Communications)*
7. Lee J*, Park J*, Sim W, Nam S, Ryu Y, **Sim H**, Kim Y, Kim SW, Song DY, Lee H, Kim W, Han JH, Lee JW, Byun HJ, Son JH, Kim SH, Oh M, Yoo HJ†, An JY†, Jeon M†, Challenges in De Novo Variant-Based Autism Spectrum Disorder Prediction Using Genomic Language Models *(Under review, Genome Biology)*
8. Do JY*, Kim T, **Sim H**, Jeong H, Choi JH, Kang M†, Optical Properties and Phenol Destruction Performance of Pd-inserted TiO_2 Photocatalysts *(Applied Chemistry for Engineering, 2017)*

ACADEMIC ACTIVITIES & HONORS

ASHG 2026 Annual Meeting (Poster; accepted)	Oct 2026
Teaching Assistant, 20th Asian Institute in Statistical Genetics and Genomics	Jul 2026
ICML 2026 GenBio Workshop (Spotlight)	Jul 2026
The 22nd KOGO Winter Symposium (Poster)	Feb 2026
Hyeongae Scholarship (full funding)	2024 – present
KU Scholarship (full funding)	2020 – 2021

SERVICE

Military Service — ROKAF 38th Fighter Group, USAF 8th Fighter Wing	2022 – 2024
Volunteer, KU Central Service Club "WoonWha"	2020 – 2022

Educational volunteering for students from low-income households (200 hours).

SKILLS

Computing: Python, R, PyTorch, HuggingFace, AWS, Slurm, Docker, Git

Data: WGS, scRNA-seq, snATAC-seq, ChIP-seq, CRISPR, MPRA, GWAS